

The USAPhO Exam

What it is, how it works, and how to write up your solutions

For official information, use the AAPT (American Association of Physics Teachers) website. If anything in this handout disagrees with that website, the website is right.

www.aapt.org/physicsteam

The pages you'll want are [Rules](#), [For Students](#), and [Past Exams](#). Everything in this handout comes from those pages or from the instructions on recent exams. The exam changes from year to year, so check there before you plan around any detail here.

1 The Short Version

The USAPhO is a three-hour, free-response exam with six problems, out of 150 points. It's given in two 90-minute halves of three problems each, with a break in between, and you can't go back to the first half after the break. You qualify by scoring above the cutoff on the $F = ma$ exam in February. The USAPhO itself is in April and is proctored in person. The questions show up on a computer screen, but you write your solutions by hand on blank paper from the proctor, and those pages get scanned and graded by people. You get a reference sheet with constants and a few approximations. You can use a basic scientific calculator. Almost nobody finishes, so most of your score comes from partial credit.

2 Part I: The Logistics

How do I qualify?

By scoring at or above the cutoff on the $F = ma$ exam. AAPT sets that cutoff after the exam, at whatever score invites roughly 400 to 500 students. Recent cutoffs have been 14 to 18 out of 25. You can't register for the USAPhO directly. See the *F = ma Exam FAQ* handout for more about that exam.

When is it?

April, a few weeks after $F = ma$ results come out. In 2026 it was April 10, inside a 1:00 to 5:00 pm EDT window (the exam is three hours plus a break, and your proctor picks the start time inside the window). Watch AAPT's "Important Dates" page. In 2026 the date moved once, from April 2 to April 10, to avoid a conflict with Passover.

What is the shape of the exam?

	Part A	Part B
Length	90 minutes	90 minutes
Problems	3 (A1, A2, A3)	3 (B1, B2, B3)
Points each	25	25
Part total	75	75

That’s 150 points total. Each problem has several lettered subparts. The instructions say that problems “may differ in difficulty” even though they’re worth the same number of points, so the order on the page doesn’t tell you which one is easiest.

Part A and Part B are separate. You can’t look at Part B during Part A, and after the break you can’t go back to Part A, since those pages have already been collected. Think of them as two separate 90-minute exams.

What topics does it cover?

All of introductory physics: mechanics, electricity and magnetism, thermodynamics and statistical physics, waves, optics, and relativity and modern physics. Unlike $F = ma$, it isn’t mechanics-only. It also expects calculus: you’ll integrate, differentiate, expand in small parameters, and solve differential equations. The 2026 exam had a geometrical optics problem, a problem about motion on a cone, a blackbody radiation problem, and a rigid-body problem, among others.

Is it on paper or on a computer?

Both. Since the 2026 cycle, AAPT says the $F = ma$ and USAPhO exams are “conducted online via Educational Vistas only,” with “no printable versions of the exams available.” Here’s what that means in practice, from the 2025 and 2026 student instructions:

- You sit at a designated desktop, laptop, or Chromebook. Phones, tablets, and smartwatches get handed in before the exam starts.
- You read the problems on the screen. When the exam begins you click the link for Part A and a 90-minute timer starts.
- You write your solutions by hand, on blank paper the proctor gives you, along with a printed instruction page, the reference sheet, and a cover sheet for each problem.
- At the end of each part the proctor collects your pages and scans them. Those scans are what get graded.

So the problems come to you online, but you’re still writing physics on paper for a person to read.

Do I get scratch paper, and how should I use it?

Yes, but only from the proctor (“Only scratch paper provided by the proctors may be used”), and it all gets collected at the end.

The rule that matters more is about what gets graded: “All work must be done on the provided blank pages... Work outside of these pages will not be graded,” and when the proctor collects your solutions, “Do not include scratch work.” The grader never sees your scratch paper.

Scratch paper is for deciding. Scratch paper is a good place to try three approaches and see which ones fall apart. Once you know which one works, you can switch to the answer pages and do the real derivation there. If you solve the whole problem on scratch paper and then copy it over, you’ve done the problem twice, and you don’t have time to do six problems twice. Even worse is grinding through a long integral on scratch paper and copying only the last line, since then you spent the time and the grader can’t see any of the work.

Is there a formula sheet?

Yes, which is a real difference from $F = ma$, where you get nothing. The reference sheet is printed in the instructions and you can use it for both parts. In 2026 it had:

- **Constants:** g , G , $k = 1/4\pi\epsilon_0$, $k_m = \mu_0/4\pi$, c , k_B , N_A , R , σ , e , 1 eV in joules, h , and m_e .
- **Approximations:** $(1+x)^n \approx 1+nx + \frac{1}{2}n(n-1)x^2$, $e^x \approx 1+x+x^2/2+x^3/6$, $\sin\theta \approx \theta - \theta^3/6$, and $\cos\theta \approx 1 - \theta^2/2$.

That’s the whole sheet. It’s really a list of constants. It doesn’t have $\frac{1}{2}mv^2$, Maxwell’s equations, moments of inertia, thermodynamic relations, or the lens equation. Some years have also had a short table of integrals and trig identities, but the 2026 sheet didn’t. You need to know your formulas.

The four approximations are a hint about what the exam likes. Expansions in a small parameter show up constantly. When a problem gives you a condition like $r \ll R$, it’s telling you which expansion to use.

What are the calculator rules?

A handheld calculator is allowed, but it “may not be a graphical calculator, its display has no more than three lines,” and you have to clear its memory of data and programs before the exam. So nothing that does symbolic math, programming, or graphing. You can’t share calculators, and the proctor won’t have one, so bring your own. You’ll use it much less than on $F = ma$, since most answers are symbolic.

What else is banned?

Books, tables, collections of formulas, notes, anything beyond the reference sheet they give you. Phones and other internet-connected devices get handed in and can’t be used at any time while exam materials are out. You have to stay in the exam room for the whole testing period.

How is it proctored?

In person and live, like $F = ma$. Proctors need at least a two-year degree, don’t need to be physics teachers, and can’t be a parent, relative, or close friend of a student taking the exam. For-profit test prep centers aren’t allowed to proctor. If you qualify, AAPT emails you and your proctor, and usually the same teacher who ran your $F = ma$ administration runs this one too.

Can I talk about the problems afterward?

Not until the embargo lifts, which is usually the day after the exam. The 2025 exam told students not to discuss any part of it until April 11, the day after. Violating that can get you disqualified. Solutions get posted publicly a few weeks later.

What happens after?

Results and an honors list come out in May. About 20 students, chosen on combined USAPhO and $F = ma$ performance, are named to the **U.S. Physics Team** and invited to a ten-day training camp at the University of Maryland in late May. AAPT's usual description is that "at the end of the camp, five students will be selected to represent the team at the International Physics Olympiad," with those five chosen from exams given during the camp.

2026 was different. A new exam, the **USAPhO Plus**, was given on May 10 to the 20 students already named to the team, and that exam, not the camp, chose the five travelers and one alternate. The camp, renamed a Training & Development Camp, ran afterward (May 30 to June 9) to prepare that delegation and to develop future competitors. The 2026 team also competed at the **European Physics Olympiad** in Gothenburg rather than the IPhO. There was no USAPhO Plus in 2024 or 2025; in 2025 the five IPhO travelers were named after camp in the usual way. It's not clear yet whether the 2026 format sticks. None of this changes what you do on exam day, but if you're in range of the top 20, read AAPT's news page for your year rather than assuming either pattern.

One thing about the 2026 USAPhO Plus is good to know about either way. Along with the usual theory problems, it had a problem with an experimental component: analyzing experimental-style data, making and interpreting plots, extracting physical parameters, and estimating uncertainties. Data analysis is also in the USAPhO syllabus, so it's a good idea to practice it.

3 Part II: How to Write Up a Solution

Start by picturing the grader

Your solution gets scanned and read by someone who has already read a few hundred versions of the same problem. They're working from a rubric that gives points for specific milestones: the right principle, the setup, an intermediate expression, the final answer. They aren't trying to reconstruct your thinking, just to find those milestones.

Most of the advice in this section comes from that.

Five habits

1. **Draw a labeled diagram** and define your symbols on it. Every angle, distance, and force you use should be visible somewhere. If the grader can't tell what your ℓ is, they can't give you credit for an equation with ℓ in it.
2. **Say which principle you're using** before you use it. One short line is enough: "torque balance about the contact point in the rotating frame," "energy conservation between release and the bottom," "Gauss's law on a cylinder of radius r ." That line usually gets points on its own, and it takes about eight seconds to write.
3. **Work with symbols** and put numbers in at the end. Symbols let the grader see the structure of what you did. A page of decimals doesn't show them anything. Also, one arithmetic slip early on messes up everything after it.

4. **Box the final answer** to every subpart. The instructions ask for this explicitly: “To help with grading, please draw a clear box around your final answer to each subpart.” An answer that isn’t boxed, sitting in the middle of the page, can get missed.
5. **One problem per sheet**, and a header on every page. Don’t mix two problems on one sheet. Label every page in the upper right.

What the page looks like

The exact answer-sheet format has changed over the years. For a long time AAPT printed framed answer sheets with an information block, and more recently you get a cover sheet for each problem plus blank paper. What hasn’t changed is the header AAPT asks for in the upper right corner of every page:

Student AAPT ID # A1 – 1/3	Proctor AAPT ID #
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That last line is the one that matters most: the problem number, then page n of N for that problem. The scans get split up and sorted by problem before grading. A page labeled only “2” can end up in the wrong pile, and a page in the wrong pile gets a zero.

A few practical rules that come from the pages being scanned and sorted:

- Write on one side only. Older instructions said so outright, and graders scan the fronts.
- Number every page as n/N , and fill in N at the end when you know it.
- Press hard and write dark. Faint pencil scans badly. Pen is safer, which is one reason crossing out beats erasing.
- Stay out of the margins. Anything in the outer half inch might get cropped.
- If you continue a problem on extra pages, say so at the bottom of the previous page (“continued on A1 – 3/3”).

What helps your score, and what hurts it

None of these are hard rules. Graders are physicists reading in good faith, and if your R is obviously the radius in your figure, nobody is going to pretend otherwise. But everything in the right-hand column makes them work harder to find the points you earned, and across a stack of three hundred papers that adds up.

Helps	Hurts
A named principle, even with no algebra after it	A page of algebra with no statement of what is being computed
A correct setup with the integral left unevaluated	A setup you did in your head, so only the result of it is on the page
A labeled diagram defining your variables	Symbols you never define, leaving the reader to guess which length is which
“I couldn’t do (a); assume the answer is X ” and then a correct (b)	A blank (b) because (a) didn’t work out
One clearly boxed answer per subpart	Three candidate answers, none crossed out
A wrong answer whose method is visible and mostly right	A wrong answer whose method is invisible

Never leave two answers standing. If you tried something, gave up on it, and tried again, put one clean line through the attempt you abandoned. Graders are generally told to grade what you present, and if you present two contradictory answers, they can grade the wrong one. It only takes a second to cross something out.

Getting past a part you can’t do

This habit helps on a lot of problems.

Subparts are usually graded independently. If part (b) says “using your result from part (a),” you don’t need part (a) to be right. You just need to have written something down for it. So:

When you’re stuck on (a) and (b) depends on it, write:

“Could not complete (a). For (b) I assume the result of (a) is $T = \dots$ ”

and then do (b) properly. You can still get full credit on (b). Two more versions of the same move:

- **Carry a symbol you can’t evaluate.** If you can’t find the moment of inertia, call it I , solve the rest of the problem in terms of I , and box that. You lose the points for I and keep the rest.
- **Do a special case.** If the general problem is too hard, solve it for the symmetric case, or to leading order in the small parameter, and say clearly that that’s what you did. That gets a lot more credit than a blank.

Two more habits that get points

Check limiting cases on the page. When you get an answer, check whether it reduces to the earlier part when the correction goes to zero, whether the sign is right, and whether the units work. Writing “as $h \rightarrow 0$ this reduces to part (a) ✓” does two things: it catches your own mistakes, and it shows a grader who is deciding between two rubric levels that you understand what you found.

Say what you're about to do before you do it. Two-word headers like “Setup,” “Torque about P ,” “Evaluate,” and “Check” turn a wall of algebra into something a reader can follow. A grader only spends a few minutes on your page, so structure helps.

4 Part III: Strategy for the Three Hours

The first five minutes: read, don't solve

Don't start on A1 the moment the clock starts. Spend the first three to five minutes reading all three problems in the part. You're looking for:

- which topic each problem is on (mechanics? thermo? circuits?),
- how many subparts each has, and whether the early subparts look doable,
- which problem is in your strongest area.

That's five minutes out of ninety. It protects you from spending forty minutes on the hardest problem of the set because it happened to be printed first, and never reading a problem you could have solved.

The order on the page isn't the order of difficulty. AAPT says so explicitly. A3 is often easier than A2.

Problem by problem, not skipping around

Once you've picked an order, work one problem at a time, start to finish. Long free-response physics isn't like multiple choice. It takes real time to reload a problem's setup into your head, so bouncing between problems costs you several minutes each time.

Within a problem, though, skip around freely. If part (b) is stuck, read (c) and (d) right away. They're often independent, or they depend on (b) in a way you can get around with an assumed result. Some of the easiest points on the exam are in the last subpart of a hard problem.

A time budget

Ninety minutes for three problems is about 30 minutes each. The hard part is actually stopping when the time is up.

Clock	Where you should be
0:00–0:05	Read all three problems. Pick an order.
0:05–0:32	First problem (your best topic).
0:32–0:59	Second problem.
0:59–1:26	Third problem.
1:26–1:30	Clean up: box unboxed answers, fix headers, cross out dead work.

Stop at 30 minutes per problem. When you hit 30 minutes on a problem, write down where you got to and box whatever partial result you have, even if you're in the middle of a derivation. If you have time at the end you can come back to it, and you'll have a fresh head. The alternative, spending 50 minutes on one problem, costs you the easy first subparts of a problem you never got to read.

A five-minute rule for a stuck subpart: if five minutes of real effort hasn't produced a new equation, it's time to write down what you know, assume a result, and go to the next subpart.

The last four minutes of each part

In the last few minutes, stop doing physics and clean up your pages:

1. Box every final answer that isn't boxed.
2. Put the problem number and n/N on every page, and fill in N .
3. Cross out every abandoned attempt so nothing contradictory is left.
4. For every subpart you left blank, write one line: the principle you would have used, the diagram you would have drawn, or the form you expect the answer to take. A sentence beats a blank, and sometimes it's a rubric line.
5. Sort your pages by problem, then by page number.

Calibrate: you're not going to get 150

It's easy to treat an unfinished problem as a failure and get rattled. But the USAPhO is designed so that even the top students in the country don't finish it.

Solving two problems completely and making real progress on two more is an excellent performance. Leaving a few subparts blank is normal. Your goal is to collect as many points as you can in 180 minutes, which is different from finishing the exam. With that goal, it makes sense to leave a hard problem, read the next one, and pick up its first two subparts.

5 Part IV: What Score Do I Need?

AAPT doesn't publish USAPhO score cutoffs, and has said it deliberately doesn't release numerical scores or detailed breakdowns. What it does publish is the *structure* of the awards, which is enough to aim by. Awards go by percentile among the students who take the exam:

Award	Share of takers	2026: about
Gold Medal	top 10–12%	top 38 students
Silver Medal	next 14–16%	next 48
Bronze Medal	next 20–22%	next 63
Honorable Mention	next 24–26%	next 79
Distinction	remainder	the rest

Above all of that is the **U.S. Physics Team**: about 20 students invited to camp, based on combined USAPhO and F = ma performance.

In 2026, 440 students qualified on F = ma and 317 actually took the USAPhO.

Two things to notice in that table. First, a medal of some color goes to roughly the top 47% of the field, and some honor goes to nearly three quarters of it. Second, the field is 300-odd people who already finished in the top 7% of F = ma takers, so the middle of this pack is a very strong physics student.

What those percentiles have meant in points

The table below comes from the actual USAPhO score files for 2014 to 2018. AAPT has never published cutoffs, so these are reconstructed the same way AAPT sets them in the first place: take that year's real distribution of scores and cut it at the published percentile bands. Through 2017 the exam was out of 200 points (four Part A problems at 25 and two Part B problems at 50); since 2018 it has been out of 150. The last block converts everything to today's 150-point scale so you can compare years.

Year	Took it	Out of	Median	Score needed for			
				HM	Bronze	Silver	Gold
2014	368	200	64	50	68	89	118
2015	363	200	67	47	71	95	129
2016	326	200	92	69	94	116	142
2017	418	200	49	32	52	71	89
2018	382	150	40	25	42	60	84
<i>On today's 150-point scale:</i>				25–52	39–70	53–87	67–106
<i>Typical year:</i>				35	50	67	88

Three things to notice.

The cutoffs move a lot. Gold ranged from 67 to 106 on today's scale. 2016 was a gentle paper (median 46% of the points) and 2017 was brutal (median 25%). So there's no fixed number to aim for. It's better to think in terms of what fraction of the exam you can do.

Everything is lower than you'd guess. In four of these five years the median was between a quarter and a third of the available points, and gold was around 55 to 60%. Even the top paper in 2018 scored 124 out of 150.

In terms of problems, on the 150-point scale, honorable mention is about a problem and a half of credit, bronze about two, silver about two and a half, and gold about three and a half out of six. None of those is a whole number of clean solutions. They're all reached by finishing some problems and picking up partial credit on others.

Where the points were in 2018. Here's the mean score on each 25-point problem, and the fraction of the field that scored zero on it:

	A1	A2	A3	B1	B2	B3
Mean score (of 25)	11.9	6.4	3.2	9.5	5.0	8.9
Scored zero	27%	10%	45%	16%	52%	15%

Half the field scored nothing at all on A3 and on B2. A3 was also the hardest problem in Part A by a wide margin, and B3 was worth nearly twice what B2 was. A student who worked strictly in order and got stuck on A3 missed the two problems in Part A that had the most points.

6 Part V: What a Real Submission Looks Like

It's easier to see all of this on an actual page. So here's one problem worked the way you'd actually work it under time pressure, including a wrong turn, followed by how it would be graded.

The problem is **2014 USAPhO, Problem A1**, which is a good one to practice on: two subparts, the second a real correction to the first, and a trap in the middle that a lot of people fall into.

Question A1 (2014 USAPhO). A unicyclist of total height h goes around a circular track of radius R while leaning inward at an angle θ to the vertical. The acceleration due to gravity is g .

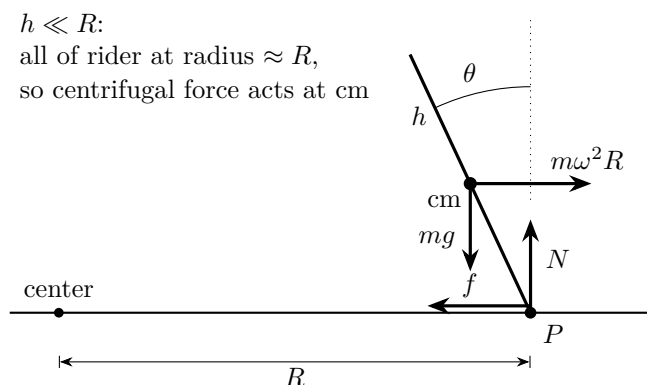
- (a) Suppose $h \ll R$. What angular velocity ω must the unicyclist sustain?
- (b) Now model the unicyclist as a uniform rod of length h , where h is less than R but not negligible. This refined model introduces a correction to the previous result. What is the new expression for the angular velocity ω ? Assume that the rod remains in the plane formed by the vertical and radial directions, and that R is measured from the center of the circle to the point of contact at the ground.

The submission

What follows is typeset so you can read it, but the content, the length, and the mess are what a good submission actually looks like. The notes in red are mine, pointing out the choices that are getting points. They wouldn't be on your page.

Student AAPT ID # Proctor AAPT ID #
A1 – 1/2

- (a) Rotating frame, rider static. Radial-vertical plane:



↑ Every symbol used below (P , θ , cm, R) is defined on the diagram. The first line says which frame we're in.

Let cm be a distance ℓ up the rod from P . Torque about P (rider is static in this frame, so $\tau_{\text{net}} = 0$; N and f act at P and contribute nothing):

$$\tau = m\omega^2 R (\ell \cos \theta) - mg (\ell \sin \theta) = 0$$

↑ The principle (torque about P) is named, and the reason N and f drop out takes half a line. That's a rubric line.

$$\omega^2 = \frac{g \tan \theta}{R}$$

$\omega^2 = g \cot \theta / R$ — lever arms swapped ← First attempt, caught and crossed out. See the check below.

$$\omega = \sqrt{\frac{g \tan \theta}{R}}$$

Check: $\theta \rightarrow 0$ (upright) $\Rightarrow \omega \rightarrow 0$. ✓ $\theta \rightarrow 90^\circ \Rightarrow \omega \rightarrow \infty$. ✓ Units: $\sqrt{(\text{m/s}^2)/\text{m}} = \text{s}^{-1}$. ✓

↑ This check is what caught the swapped lever arms. It took maybe thirty seconds, and it's the difference between 8/8 and 3/8 on this part.

Student AAPT ID # Proctor AAPT ID #
A1 – 2/2

(b) Now $h \not\ll R$, so different parts of the rod are at different radii and the centrifugal force is *not* uniform.

Idea: cm sits at radius $R - \frac{h}{2} \sin \theta$, so replace $R \rightarrow R - \frac{h}{2} \sin \theta$ in (a).

← The trap, tried and abandoned. There's a one-line reason written next to it instead of a silent erasure:

No - torque involves the product rz , and $\langle rz \rangle \neq \langle r \rangle \langle z \rangle$. Must integrate.

Setup. Mass element a distance s along the rod from P :

$$r = R - s \sin \theta, \quad z = s \cos \theta, \quad dm = \frac{m}{h} ds, \quad 0 \leq s \leq h$$

↑ Each of these three is a rubric line on its own. Even if the integral had gone nowhere, these already get points.

Centrifugal torque about P . Force $\omega^2 r dm$ outward, lever arm z :

$$\begin{aligned}\tau_c &= \int_0^h \omega^2 (R - s \sin \theta) (s \cos \theta) \frac{m}{h} ds \\ &= \frac{m\omega^2 \cos \theta}{h} \left[\frac{Rh^2}{2} - \frac{h^3 \sin \theta}{3} \right] = m\omega^2 h \cos \theta \left(\frac{R}{2} - \frac{h \sin \theta}{3} \right)\end{aligned}$$

Gravity torque about P , at the cm, $h/2$ up the rod:

$$\tau_g = -mg \frac{h}{2} \sin \theta$$

Balance. $\tau_c + \tau_g = 0$:

$$\omega^2 h \cos \theta \left(\frac{R}{2} - \frac{h \sin \theta}{3} \right) = g \frac{h}{2} \sin \theta \quad \implies \quad \omega^2 = \frac{g \tan \theta}{R - \frac{2h}{3} \sin \theta}$$

$$\boxed{\omega = \sqrt{\frac{g}{R} \tan \theta \left(1 - \frac{2h}{3R} \sin \theta \right)^{-1}}}$$

Checks: $h \rightarrow 0$ reduces to (a) ✓ Denominator $< R$, so ω is *larger* than in (a) ✓ - correct, since the rod's mass sits at smaller radius than R and so needs a faster spin to hold the lean.

↑ The first check shows the correction is consistent with (a). The second shows the sign of the correction was understood, not just stumbled into.

How that gets graded

USAPhO rubrics aren't published, so the breakdown below is a reconstruction, but it has the right shape: a 25-point problem is cut into five or six milestones, and each one is awarded on its own.

Rubric line	Pts	Earned
(a) Point-mass limit		
Correct force picture: states the frame and identifies the centrifugal force acting at the cm	2	2
Correct torque (or force) balance about the contact point	3	3
Correct $\omega = \sqrt{g \tan \theta / R}$	3	3
(b) Extended rod		
Recognizes the centripetal acceleration varies along the rod, so the cm substitution fails	3	3
Correct parametrization $r = R - s \sin \theta$, $z = s \cos \theta$, $dm = (m/h)ds$	4	4
Correct integral expression for the centrifugal torque	4	4
Correct evaluation $m\omega^2 h \cos \theta \left(\frac{R}{2} - \frac{h \sin \theta}{3} \right)$	3	3
Correct gravitational torque and balance	3	3
Correct final ω , boxed	3	3
Total	25	25

That submission gets full credit, even though very little of the page is polished. There's a crossed-out formula, an abandoned idea, and hardly any prose. It gets a 25 because every rubric line is easy to find.

The same physics, written four other ways

All four of these students did roughly the same amount of physics.

What the page contains	Score
Ran out of time in (b). Part (a) complete. In (b): "the centripetal acceleration varies along the rod, so I have to integrate the torque instead of using the cm," plus $r = R - s \sin \theta$, $z = s \cos \theta$, $dm = (m/h)ds$, and the integral written down but never evaluated.	18/25
Fell into the trap. Part (a) complete. In (b), substituted $R \rightarrow R - \frac{h}{2} \sin \theta$ with no comment and boxed it.	9/25
Blew part (a), recovered. Part (a) has the lever arms swapped and is never checked, so $\cot \theta$ is boxed. Part (b) is done correctly from scratch, with the setup, integral, evaluation, balance, and answer all there.	22/25

What the page contains	Score
Did all the physics, wrote none of it. Everything correct, but done on scratch paper, with only the two boxed answers copied onto the answer sheets.	6/25

The third student got *part (a) wrong* and still scored 22, since subparts are graded independently and (b) was written out. The fourth student got everything right but scored 6 because the work was on scratch paper. The difference between 6 and 22 here has nothing to do with physics.

Things that go wrong, and what to do about them

You suspect part (a) is wrong but can't see why.

Try the limiting cases, which is what caught the swapped lever arms above. Set each parameter to zero or infinity and see whether the answer still makes sense. If a check fails, you've found a real error. If they all pass, there's no reason to keep worrying about it, especially since a wrong (a) doesn't sink (b) anyway.

You can't see how to set up the integral in (b).

Write down what you *do* know ("the acceleration varies along the rod, so $\tau_c = \int \omega^2 r z dm$ "), then define r , z , and dm one at a time from your diagram. That setup alone gets 7 of the 17 points in (b), before you've integrated anything.

You get an ugly answer and think you must have made a mistake.

Ugly answers are normal. $\omega^2 = g \tan \theta / (R - \frac{2h}{3} \sin \theta)$ is ugly, and it's correct. If it passes the limiting cases, it's probably fine, and there's no reason to spend ten minutes hunting for a mistake just because you expected something prettier.

You have two versions of an answer and no time to decide.

Pick the one that passes your limiting checks and cross out the other one.

You're 30 minutes in with (b) half done.

Box whatever you have and write one line saying what's left ("still need to evaluate the integral and balance against gravity"). Then go read the next problem. Its first subpart is probably worth more points than your next ten minutes on this one.

7 Part VI: Two Real Papers

The example above was made up. These two are real 2018 USAPhO submissions to Problem B2, reproduced as the graders received them, with the scores they actually got. The scans only carry the AAPT numbers printed on the answer sheet, so there are no names on them.

Question B2 (2018 USAPhO). In this problem, use a particle-like model of photons: they propagate in straight lines and obey the law of reflection, but are subject to the quantum uncertainty principle. You may use small-angle approximations throughout.

A photon with wavelength λ has traveled from a distant star to a telescope mirror, which has a circular cross-section with radius R and a focal length $f \gg R$. The path of the photon is nearly aligned to the axis of the mirror, but has some slight uncertainty $\Delta\theta$. All small cross-sectional areas of the mirror are equally likely to include the point of reflection.

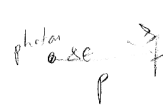
- (a) Find the standard deviation σ_r of the distribution for r , the distance from the center of the mirror to the point of reflection.
- (b) Use the uncertainty principle, $\sigma_r \sigma_{p_r} \geq \hbar/2$, to place a bound on how accurately we can know the angle of the photon from the axis of the telescope. Give your answer in terms of R and λ . *If you were unable to solve part a, you may also give your answer in terms of σ_r .*
- (c) Suppose we want to tell with high probability whether a detected photon came from the left half or right half of Alpha Centauri A. Approximately how large would the telescope have to be? The star is about 4×10^{16} m away with radius about 7×10^8 m; assume $\lambda = 500$ nm.

Notice the italicized sentence in part (b). The exam is telling you that you can use the move from Part II. Only one of these two students took the offer.

Paper 1: 22 out of 25

a) Variance $= \overline{r^2} = \frac{\int_0^R 2\pi r \cdot r^2 dr}{\pi R^2} = \frac{1}{2} R^2$ #4717
#69012

$\hookrightarrow \Delta r = \frac{R}{\sqrt{2}}$ B2-1/1

b)  $p = \frac{h}{\lambda} \rightarrow \Delta p_r = p \Delta \theta = \frac{h}{\lambda} \Delta \theta$

$\hookrightarrow \frac{R}{\sqrt{2}} \cdot \frac{h}{\lambda} \Delta \theta \geq \frac{h}{2}$

$\Delta \theta \geq \frac{1}{\sqrt{2}} \cdot \frac{\lambda}{R}$

c) We want $\Delta \theta \leq \frac{7 \cdot 10^8}{4 \cdot 10^{16}} = 1.75 \cdot 10^{-8}$

Rearranging above,

$R \geq \frac{1}{\sqrt{2}} \cdot \frac{\lambda}{\Delta \theta}$

$= \frac{1}{\sqrt{2}} \cdot \frac{5 \cdot 10^{-7}}{1.75 \cdot 10^{-8}}$

$\boxed{\approx 20 \text{ m.}}$

What they did. All of B2 on one side of one sheet, headed B2 – 1/1. In (a) they wrote the area-weighted average straight down, $\int_0^R 2\pi r \cdot r^2 dr / \pi R^2 = \frac{1}{2} R^2$, and took the square root. In (b) they sketched the photon and the angle, wrote $p = h/\lambda$ and $\Delta p_r = p \Delta \theta$, put those into the uncertainty relation, and boxed $\Delta \theta \geq \frac{1}{\sqrt{2}} \lambda/R$. In (c) they got the angle the telescope has to resolve, $7 \times 10^8 / 4 \times 10^{16} = 1.75 \times 10^{-8}$, rearranged their own (b) for R , and boxed about 20 m.

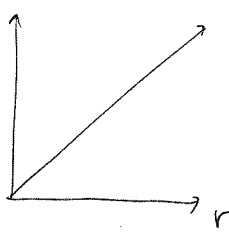
Why they got the credit they did. Each subpart is a short chain a grader can follow start to finish in a few seconds, and both final answers are boxed. Part (a) is actually wrong: they computed the root-mean-square radius, not the standard deviation. With $\langle r \rangle = 2R/3$, the variance is $\langle r^2 \rangle - \langle r \rangle^2 = R^2/18$, so $\sigma_r = R/(3\sqrt{2})$, smaller than what they wrote by a factor of three. They lost the points for (a) and kept nearly everything else, because (b) and (c) were carried through correctly *from their own* (a). This is normal. Graders give credit for a later part done correctly with a wrong number from an earlier part.

What would have earned more. Only that one thing. Writing down the definition they were

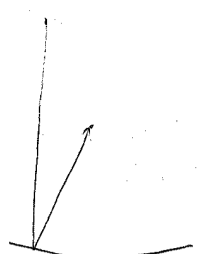
using, $\sigma^2 = \langle r^2 \rangle - \langle r \rangle^2$, before computing anything would probably have caught the mistake, since the missing $\langle r \rangle^2$ term is visible as soon as the formula is on the page. Naming the principle first is also how you catch your own mistakes.

Paper 2: 5 out of 25

a)



b)



$p_r = \frac{h}{\lambda}$

c)

$\Delta\theta < \tan^{-1}\left(\frac{7}{4} \times 10^{-8}\right)$

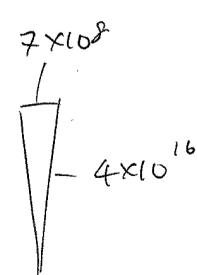
$\Delta\theta < 1.75 \times 10^{-8}$

Student # 4900

Proctor # 47982

Problem B2

Page 1 of 1



The same student scored 25/25 on A1 and 23/25 on A3 that day, and A3 was the problem where 45% of the field scored zero. So this student knew plenty of physics.

What they did. In (a), a pair of axes labeled r with an arrow across them, and no distribution, integral, or answer. In (b), a sketch of a photon hitting the mirror and a single equation, $p_r = h/\lambda$. In (c), a sketch of the star and the angle it subtends at Earth, and $\Delta\theta < \tan^{-1}(\frac{7}{4} \times 10^{-8}) = 1.75 \times 10^{-8}$.

Why they got the credit they did. Two of the three physical ingredients on that page are right, and they're the same two Paper 1 used: the photon momentum h/λ , and the angular size of Alpha Centauri A. That's what the 5 points are for. But nothing on the page connects them. Part (b) never uses the uncertainty principle, so there's no bound on $\Delta\theta$. Part (c) finds the angle the telescope has to resolve and stops there, without ever getting to the size of the telescope, which is what the question asked for. Nothing is boxed.

What would have earned more. Three lines, none of which needed a new idea:

- **Take the offer in part (b).** The problem says outright that you can answer in terms of σ_r . Writing $\sigma_r \cdot \frac{h}{\lambda} \Delta\theta \geq \hbar/2$ and solving for $\Delta\theta$ needs nothing from part (a), and $p = h/\lambda$ was already written down two inches above.
- **Finish (c) with what you have.** Even with no bound from (b), $R \sim \lambda/\Delta\theta = 5 \times 10^{-7}/1.75 \times 10^{-8} \approx 30$ m is one division, and it answers the question. An order-of-magnitude answer gets credit, but an intermediate quantity doesn't get much.
- **Put a sentence next to each sketch.** Three of the drawings on this page have nothing written next to them. The (c) sketch has exactly the right idea in it (the star's radius over its distance is the angle you need to resolve), but a grader can only give credit for reasoning they can identify. A few words next to the sketch would have done it.

Both students had the same two facts on the page. One turned them into two boxed numbers and scored 22, and the other left them unconnected and scored 5. The difference wasn't how much physics they knew.

Paper 2 isn't unusual. B2's mean score was 5.0 out of 25, and 52% of the field scored zero on it, even though the exam had told everyone, right in part (b), how to get past the step that stopped them.

8 Checklist

Before the exam

- ☐ Confirmed the date, start time, and site with your proctor
- ☐ Non-graphing scientific calculator, memory cleared
- ☐ Memorized the formulas the reference sheet doesn't give you (it's constants and four approximations, nothing else)
- ☐ Practiced on at least two past USAPhOs under timed conditions, writing on paper, in 90-minute blocks

During the exam

- ☐ Read all three problems before starting any of them
- ☐ Diagram first, symbols defined on it
- ☐ Name the principle before the algebra
- ☐ Symbols throughout; numbers only at the end
- ☐ Box every subpart's final answer
- ☐ Header on every page: problem number and n/N
- ☐ Solve on the answer sheets, not on scratch paper

- ☐ Stuck? Write down an assumed result and do the next part anyway
- ☐ 30 minutes per problem, then stop
- ☐ Last four minutes: box, label, cross out, sort

If you have questions this handout doesn't answer, ask me. And check aapt.org/physicsteam, which always has the final say.